

FRANCISCO LOURENÇO

3D Data Specialist | ML & CV Engineer

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SUMMARY

Versatile and passionate Machine Learning Engineer with strong expertise in Computer Vision, 3D data processing, and end-to-end system development. Experienced in designing, building, and deploying production-grade ML solutions across diverse domains, from industrial automation to large-scale 3D applications. Skilled in leading research-driven projects from prototyping to deployment, blending creativity, technical depth, and practical engineering. Thrives in fast-paced, collaborative environments and is highly motivated to tackle challenging problems that require innovative and reliable AI solutions.

WORK EXPERIENCE

3D Data Specialist | ML & CV Engineer

Freelance

- Apr 2025 – Present Coimbra, Portugal (Remote)
- Developed a LiDAR-camera colorization pipeline with artifact filtering for high-quality RGB point clouds.
 - Implemented ortho-photo stitching (“orthostrips”) with rectification, seam-finding, and spatial metadata.
 - Built georeferencing modules aligning LiDAR, imagery, and GPS/INS via quaternion interpolation and trajectory refinement.
 - Developed a 3D reconstruction pipeline for houses using geospatial data and aerial imagery.

Machine Learning Engineer

Anything World

- Jan 2023 – Apr 2025 London, UK (Remote)
- Designed and deployed ML pipelines for 3D shape classification, canonical alignment, and part segmentation used in the Animate Anything tool and Anything World API.
 - Achieved 95.6% F1-score on the rotated ModelNet40 benchmark through a customized multi-view GCN architecture optimized for rotation invariance and computational efficiency.
 - Developed a proprietary 3D rotation estimation system (Symmetry-Aware and Base-Plane-Aided) achieving 2.47° median angular error across 220+ characters.
 - Built an optimized 3D shape segmentation pipeline with a GCN model reaching 88.8% mIoU on ShapeNet Part; reduced training and inference time significantly compared to the prior system.

3D Computer Vision Engineer

coatingAI

- Sep 2021 – Jan 2023 Barcelona, Spain
- Solely designed and developed a multi-camera RGB-D 3D scanning system for automated part inspection in powder coating lines.
 - Built and deployed a full pipeline for 3D reconstruction, camera pose estimation, and point cloud fusion from stereo and depth data.

Junior Researcher

Institute of Systems and Robotics

- Jan 2021 – Jun 2021 Coimbra, Portugal
- Developed an anomaly detection system for glass bottle production lines, combining classical computer vision and machine learning methods.
 - Worked with real-world industrial datasets under high-throughput and high-accuracy inspection constraints.

EDUCATION

MSc in Electrical and Computer Engineering

University of Coimbra

- 2015 – 2021 Portugal
- Specialized in Computer Vision and Machine Learning.
 - Thesis: *6DoF Object Pose Estimation from RGB-D Images using ML* — graded 19/20.

MSc Exchange Program

Aalto University

- 2018 – 2019 Espoo, Finland
- Coursework focused on AI and Robotics.

PUBLICATIONS

Intel RealSense SR305, D415 and L515:
Experimental Evaluation and Comparison of
Depth Estimation
VISIGRAPP 2021

Evaluation of the Accuracy of Pose
Estimation Based on Relative Pose
RECPAD 2021

TECHNICAL SKILLS

- Python PyTorch TensorFlow OpenCV
Open3D C++ SQL HTML CSS
MatLab ROS Blender CAD Docker

FIELDS OF INTEREST

- Machine Learning Computer Vision
Point Clouds Data Science Mathematics
Coding Health Care Robotics
Automation and Control
Self Driving Vehicles 3D Animation